STA 6222

Crash Unit A

TB Models + Malaria Models

Goal: cover disease-specific models fast, but properly.
Focus: model structure, parameter meaning, thresholds, and intervention levers.

Learning outcomes

What you should be able to do after this crash unit

By the end you can...

- Write a minimal TB latency model (S-L-I) and explain why latency changes persistence.
- Add reinfection and treatment to TB models; describe a basic MDR-TB extension.
- Write a Ross-Macdonald type malaria model (humans + vectors) and interpret key parameters.
- Explain how ITNs, IRS, larval control, and treatment modify parameters and reduce transmission.
- Introduce seasonality and describe its impact on observed incidence patterns and fitting.

Roadmap

Two diseases, one modelling mindset

Part 1: TB

- Latency (L) is the whole story
- Progression vs clearance
- Reinfection and treatment
- MDR as a two-strain extension

Part 2: Malaria

- Vectors mediate transmission
- Ross-Macdonald structure
- R0 and intervention levers
- Seasonality (rainfall/temperature)

TB modelling: what makes it special

Key epidemiologic features

- Long latency: infection does not mean immediate infectiousness.
- Slow time scales: persistence even when short-term incidence falls.
- Reinfection possible (partial immunity), especially in high-burden settings.
- Treatment is central (case finding + linkage + completion).
- Drug resistance introduces a second “problem epidemic”.

TB: minimal compartment structure

S → L → I (and optionally T)

$$S \text{ (Susceptible)} \xrightarrow[\text{infection}]{(\lambda S)} L \text{ (Latent)} \xrightarrow[\text{progression}]{(\kappa L)} I \text{ (Infectious)} \xrightarrow[\text{treatment}]{(\tau I)} T \text{ (On treatment)}$$

Big idea: latency adds memory — today's infections can become tomorrow's cases.

TB: minimal S–L–I equations

Force of infection and core dynamics (illustrative, closed population):

$$\lambda(t) = \beta \frac{I(t)}{N}$$

$$\frac{dS}{dt} = -\lambda S, \quad \frac{dL}{dt} = \lambda S - \kappa L, \quad \frac{dI}{dt} = \kappa L - \gamma I$$

Parameter meanings

- β: transmission rate; κ: progression/reactivation; γ: removal from infectious state.
- Average latent duration ≈ $1/\kappa$; average infectious duration ≈ $1/\gamma$.

TB: a useful heuristic threshold

Progression probability matters

If latent infection can clear/exit without progressing (rate ω):

$$R_0 = \frac{\beta}{\gamma} \cdot \frac{\kappa}{\kappa + \omega}$$

Interpretation cues

β/γ: "infectiousness × infectious duration".

- β/γ: "infectiousness × infectious duration".

- $\kappa/(\kappa+\omega)$: fraction of new infections that eventually become infectious TB.
- Control can target either side: reduce β or shorten infectious duration ($\uparrow\gamma/\tau$), AND reduce progression ($\downarrow\kappa$ or $\uparrow\omega$).

Control can target either side: reduce β or shorten infectious duration ($\uparrow\gamma/\tau$), AND reduce progression ($\downarrow\kappa$ or $\uparrow\omega$).

TB: reinfection (partial immunity)

Allow latently infected people to be reinfected at relative susceptibility ρ (0–1):

$$\frac{dL}{dt} = \lambda S - \kappa L + \rho \lambda L$$

What changes when ρ increases?

- More “recycling” through infection → higher persistence and slower decline.
- Treatment alone may not be enough in high-exposure settings.
- Makes infection control and prevention (crowding/ventilation, preventive therapy) more valuable.

TB: treatment as a programmatic lever

Simple treatment representation:

$$\frac{dI}{dt} = \kappa L - (\gamma + \tau)I, \quad \frac{dT}{dt} = \tau I - \delta T$$

Translate to real-world actions

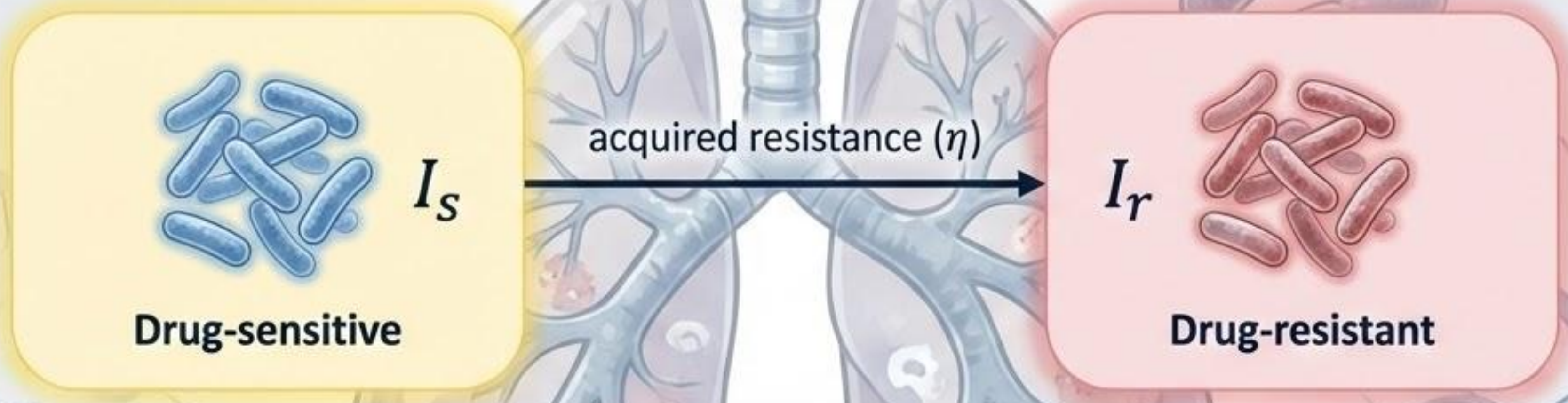
τ (treatment initiation): active case finding, better diagnostics, faster linkage.

- $\uparrow \tau$ (treatment initiation): active case finding, better diagnostics, faster linkage.
- $\uparrow \delta$ (completion/cure): adherence support, drug supply reliability, patient-centered care.
- $\downarrow \beta$ (transmission): ventilation, masks in clinics, reducing overcrowding.

$\downarrow \beta$ (transmission): ventilation, masks in clinics, reducing overcrowding.

TB: MDR as a two-strain extension

Split infectious TB into drug-sensitive (I_s) and drug-resistant (I_r).



Key implications

- If I_r has lower treatment success, infectious duration increases → resistant transmission grows.
- You can “win” against I_s and still lose overall if I_r expands.
- Control levers: rapid DST, effective MDR regimens, preventing acquisition during treatment.

TB: what data estimate what?

Incidence, Diagnosis, & Treatment (β, τ, δ)

β: inferred by fitting incidence trends (sensitive to contact patterns and reporting).

τ: informed by diagnosis-to-treatment delays + program intensity (case finding).

δ: treatment outcomes (completion/cure) from registers.

Latency & Reactivation (κ, ω)

κ, ω: best informed by cohort/longitudinal studies;

often calibrated rather than directly estimated.

Resistance parameters (η, δ_r):

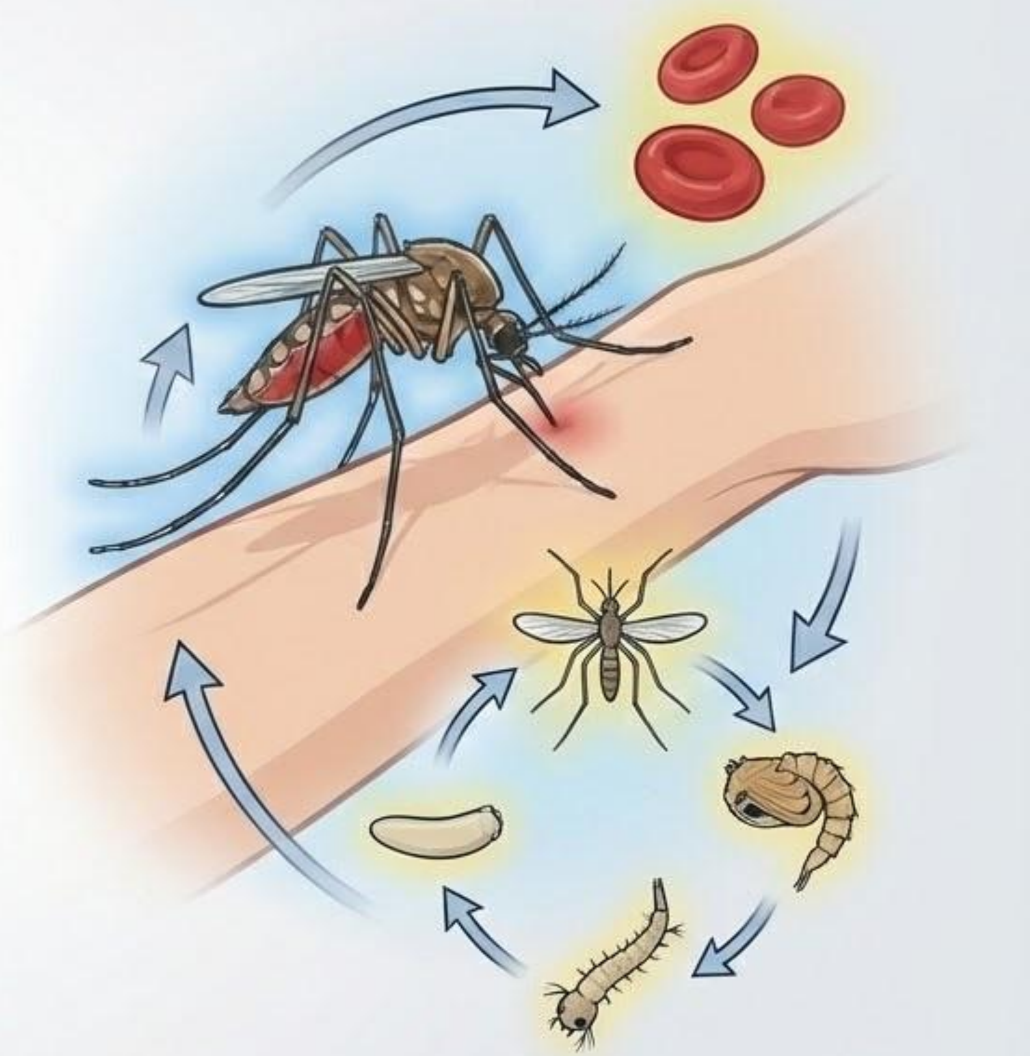
treatment failure/relapse + lab resistance data.

Part 2: Malaria models

Malaria models

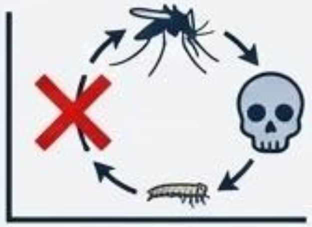
Vectors drive transmission.

Control the vector, you control the system.

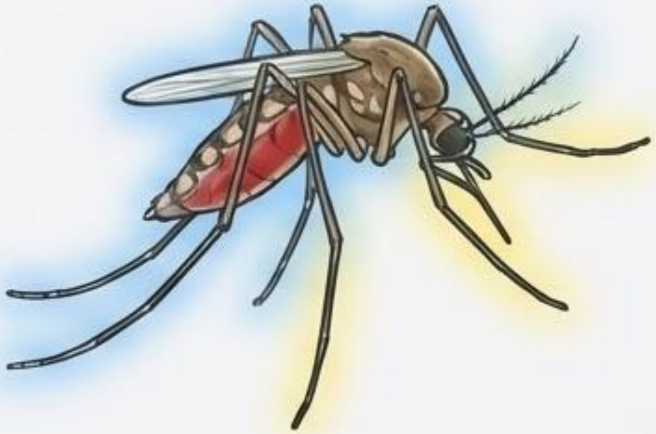


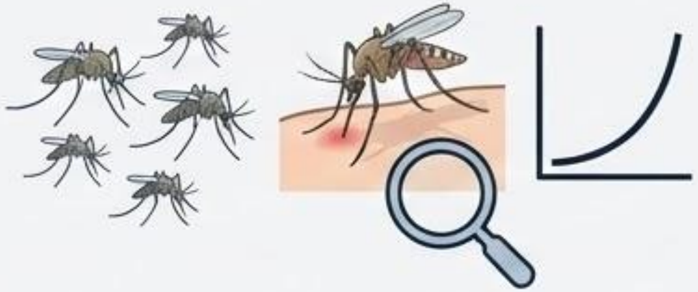
Malaria: why vectors must be in the model

Transmission is human → mosquito → human (not human → human).



Mosquito survival is critical: if mosquitoes die early, transmission collapses.





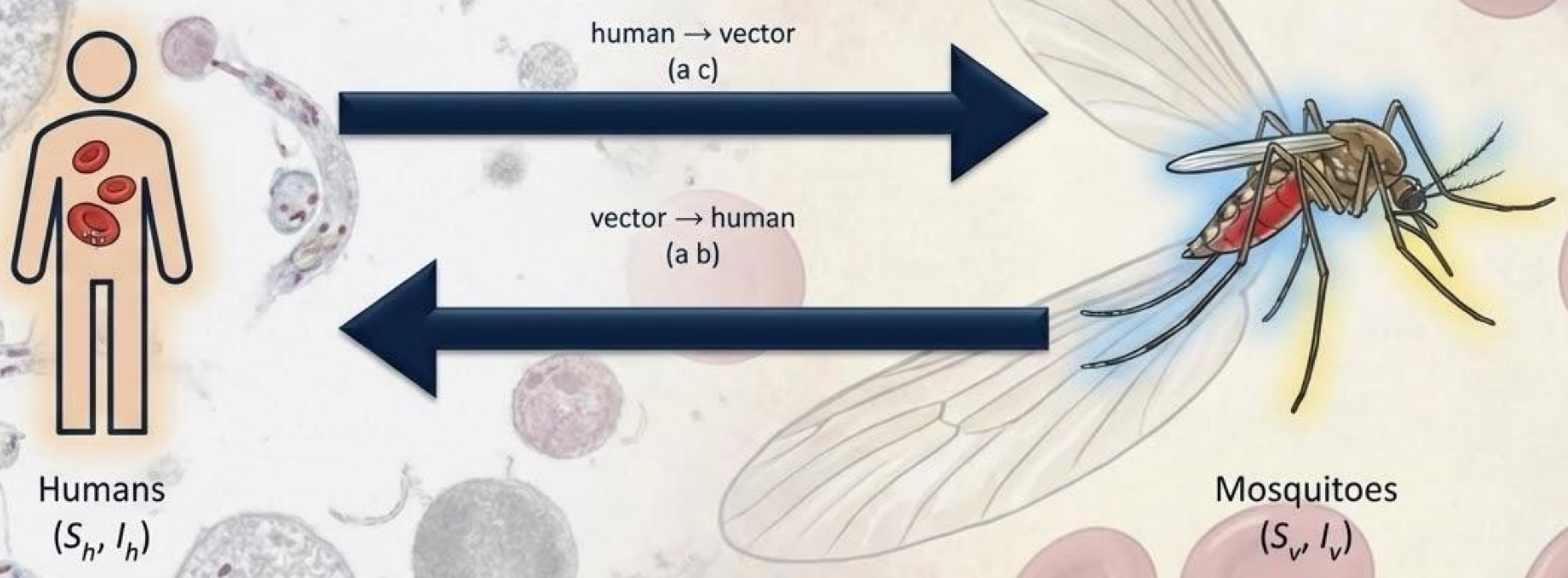
Vector density and biting rate create strong non-linear leverage for control.

Seasonality often enters through vector density and biting (rainfall, temperature).

Host–vector transmission loop

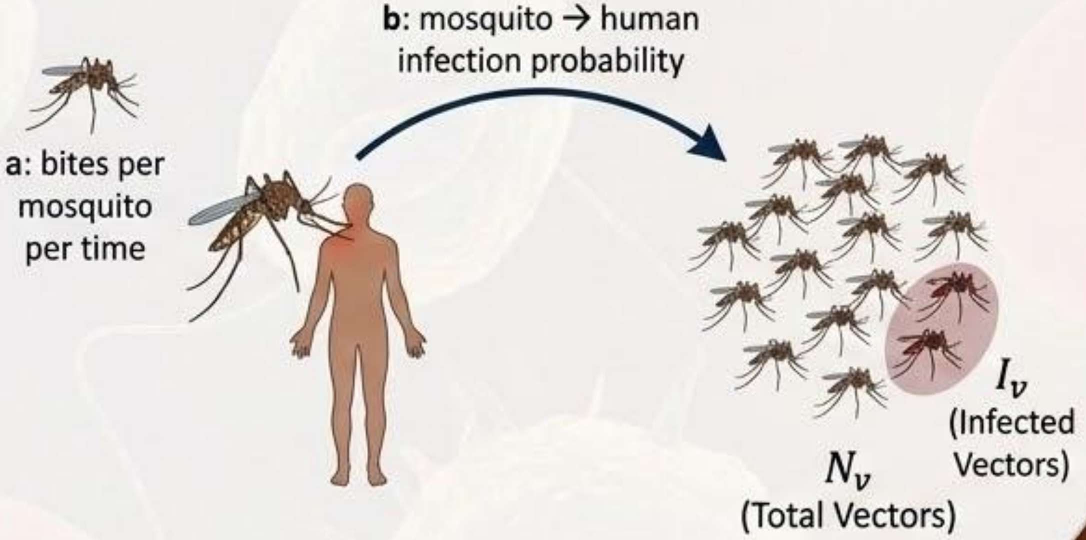
Key takeaway

- Two forces of infection: λ_h depends on I_v; λ_v depends on I_h.

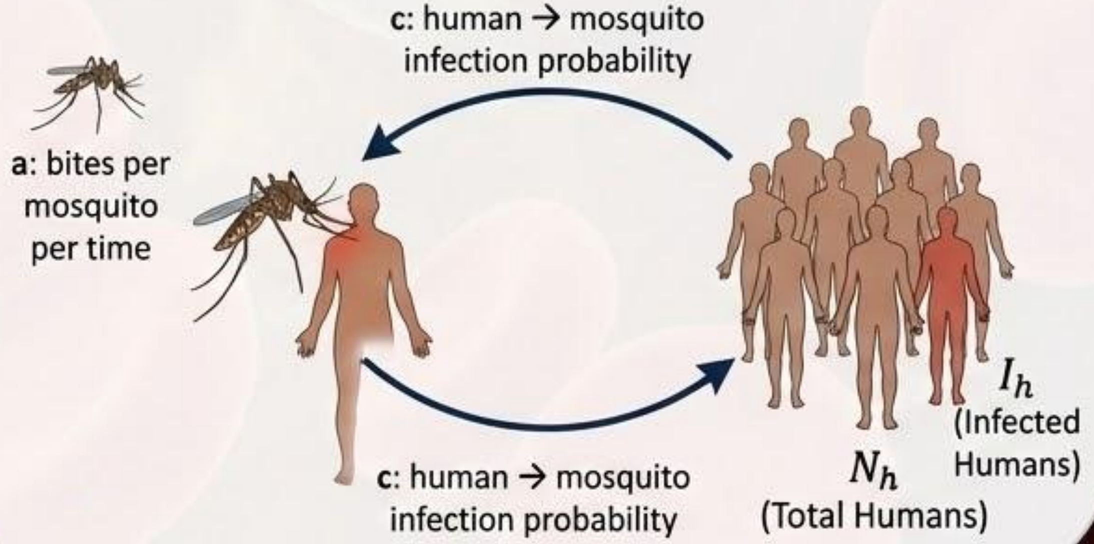


Malaria: forces of infection

$$\lambda_h = ab\frac{I_v}{N_v}$$



$$\lambda_v = ac\frac{I_h}{N_h}$$



Vector control mostly attacks a, μ_v, and the vector density m.

Malaria: minimal infectious dynamics

$$\frac{dI_h}{dt} = \lambda_h S_h - \gamma_h I_h \qquad \frac{dI_v}{dt} = \lambda_v S_v - \mu_v I_v$$

Two rates matter a lot

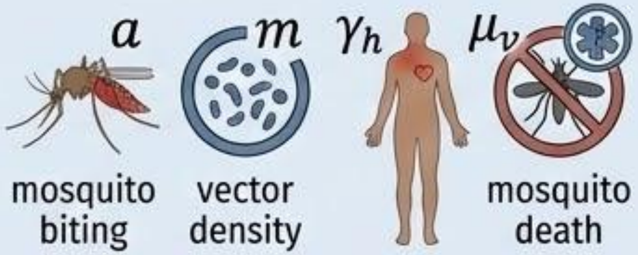
γ_h: human recovery
(treatment increases γ_h).

μ_v: mosquito death
(IRS increases μ_v; ITNs can
increase μ_v via insecticide).

Malaria: threshold (R0) and interpretation

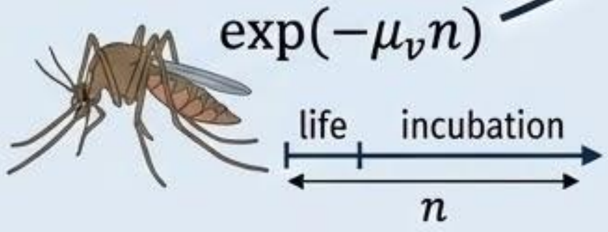
$$R_0 = \sqrt{\frac{a^2 b \cdot c \cdot m}{\gamma_h \mu_v}} \exp(-\mu_v n)$$

What to say in an exam



Parameter Influences

R_0 increases with a and m; decreases with γ_h and μ_v.



Survival & Incubation

$\exp(-\mu_v n)$ is survival through incubation: short mosquito lifespan kills transmission.

Vector Control's Power

Vector control is powerful because it attacks multiple multiplicative terms.

Interventions as parameter changes

Vector-focused

ITNs: $\downarrow a$ and often $\uparrow \mu_v$

IRS: $\uparrow \mu_v$

Larval control: $\downarrow m$

Housing/behavior: $\downarrow a$ (exposure)

Human-focused

Prompt diagnosis/treatment: $\uparrow \gamma_h$

Chemoprevention: lowers effective susceptibility

Vaccines (where used): lower infection or disease probability

Rule-of-thumb

If $R0 \propto a \cdot \sqrt{\frac{m}{\mu_v}}$ (holding other terms constant):

Reducing a by 30% (ITNs):
Reduces R0 by 30%. $(1 - 0.30) = 0.7$

Increasing μ_v by 50% (IRS):
Reduces R0 by about $1/\sqrt{1.5} \approx 0.816$ (≈18% reduction)

Together (Synergy):
$0.7 \times 0.816 \approx 0.57$ (≈43% reduction in R0)

Seasonality

Model it as a time-varying parameter

Example (biting rate varies through the year):

$$a(t) = a_0\left[1 + \alpha \sin\left(\frac{2\pi t}{365}\right)\right]$$

Practical implications

Incidence peaks may lag rainfall (vector build-up + incubation + reporting delays).

- Incidence peaks call (vector build-up + incubation + reporting delays).
- Fitting becomes harder: and interventions can be confounded.
- Use covariates (rairature) and intervention coverage data where possible.

Use covariates (rainfall/temperature) and intervention coverage data where possible.

Malaria: what data estimate what?

Practical estimation cues

Human surveillance:
Weekly cases, test positivity → informs λ_h and γ_h (via treatment).

Entomology:
Mosquito density, sporozoite rate, EIR → informs a, m, and transmission probabilities.

Mosquito survival:
Parity rates or mark–release–recapture → informs μ_v.

Coverage data:
ITN use, IRS coverage → maps to changes in a and μ_v.

TB vs Malaria (compare-and-contrast)

TB

- Latency (L) drives persistence
- Reinfection possible
- Treatment central; MDR matters

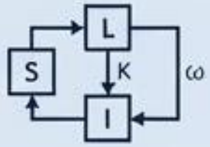
Malaria

- Vectors mediate transmission
- Mosquito survival is crucial
- Strong leverage from ITNs/IRS
- Strong seasonality

In both: model = biology + data + interventions.
Always explain parameters and what your assumption buys (and costs).

Exam-style prompts (practice thinking)

If you can answer these, you're safe



Write the TB S–L–I model; explain κ, γ, ω and how preventive therapy changes them.

Explain how reinfection (ρ) changes persistence and control thresholds.

Write a host–vector model; define a, b, c, m, μ_v, γ_h.

Explain how ITNs/IRS change parameters and why mosquito survival matters ($\exp(-\mu_v n)$).

Add seasonality to one parameter and explain how it changes observed incidence and fitting.

Problem set

Use it as the “second lecture”

How to use the problem set

Attempt questions without looking at solutions first.

Then use the full solutions to check method and interpretation.

Focus on: writing correct equations, defining parameters, and making clear intervention arguments.

Summary

Crash Unit A in 6 lines

Tuberculosis (TB)

- TB: latency creates persistence; control targets transmission, progression, and infectious duration.
- Reinfection and MDR increase the control challenge (more pathways to sustain transmission).

Malaria

- Malaria: transmission is vector-mediated; mosquito survival is a key bottleneck.
- Vector control (ITNs/IRS/larval) attacks multiplicative terms → big impact.

Modeling Principles

- Seasonality is real: put it into parameters, and expect lags.
- Always link interventions to parameter changes and state assumptions clearly.

Questions?

Bring your problem-set sticking points.